GOVRNMENT POLYTECHNIC
JAMNAGAR

MICRO PROJECT

ON

“ATM SIMULATOR”

SUBJECT: ADVANCED OBJECT-ORIENTED PROGRAMMING

CODE:4340701

SUBMITED BY

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DEFINATION: -

This Java micro-project is an ATM Simulator,

designed to mimic basic ATM functionalities. It

allows users to:

The project uses file handling to store and retrieve

balance data, an ArrayList to maintain transaction

history, and multi-threading to log transactions

separately. It ensures user-friendly interaction

with error handling and smooth execution.

Source Code:-

```
import java.io.*;
import java.util.ArrayList;
import java.util.Scanner;
```

```java
interface ATMOperations
{
    void checkBalance();
    void deposit(double amount);
    void withdraw(double amount);
    void showTransactionHistory();
}

class ATM implements ATMOperations
{
    private double balance;
    private ArrayList<String> transactions;
    private final String fileName = "balance.txt";

    public ATM()
    {
        this.balance = balancedata();
        this.transactions = new ArrayList<>();
    }

    private double balancedata()
    {
        try (BufferedReader reader = new BufferedReader(new FileReader(fileName)))
        {
            return Double.parseDouble(reader.readLine());
        } catch (IOException | NumberFormatException e)
        {
            System.out.println("Error reading balance file. Setting default balance ₹5000.");
            return 5000;
        }
    }

    private void saveBalance()
    {
        try (BufferedWriter wtr = new BufferedWriter(new FileWriter(fileName)))
        {
            wtr.write(String.valueOf(balance));
        } catch (IOException e)
        {
            System.out.println("Error saving balance to file.");
        }
    }

    public ArrayList<String> getTransactions()
    {
        return transactions;
    }

    public void checkBalance()
    {
        System.out.println("Current balance: ₹" + balance);
    }

    public void deposit(double amount)
    {
        if (amount > 0)
        {
            balance += amount;
            transactions.add("Deposited ₹" + amount);
            saveBalance();
            System.out.println("₹" + amount + " deposited successfully.");
```

```
        } else
        {
            System.out.println("Invalid deposit amount.");
        }
    }

    public void withdraw(double amount)
    {
        if (amount > 0 && amount <= balance)
        {

            balance -= amount;
            transactions.add("Withdrawn ₹" + amount);
            saveBalance();
            System.out.println("₹" + amount + " withdrawn successfully.");
        } else
        {
            System.out.println("Invalid withdrawal amount or insufficient balance.");
        }
    }

    public void showTransactionHistory()
    {
        if (transactions.isEmpty())
        {
            System.out.println("No transactions yet.");
        } else
        {
            System.out.println("\nTransaction History:");
            for (String transaction : transactions)
            {
                System.out.println(transaction);
            }
        }
    }
}

class TransactionLogger extends Thread
{
    private ArrayList<String> transactions;

    public TransactionLogger(ArrayList<String> transactions)
    {
        this.transactions = transactions;
    }

    public void run()
    {
        try (BufferedWriter wtr = new BufferedWriter(new FileWriter("transactions.log", true)))
        {
            for (String transaction : transactions)
            {
                wtr.write(transaction + "\n");
            }
            System.out.println("Transaction log updated.");
        } catch (IOException e)
        {
            System.out.println("Error writing transaction log.");
        }
    }
}
```

```
public class ATMSimulator
{
    public static void main(String[] args)
    {
        Scanner scanner = new Scanner(System.in);
        ATM atm = new ATM();

        while (true)
        {
            System.out.println("\nATM Menu:");
            System.out.println("1. Check Balance");
            System.out.println("2. Deposit Money");
            System.out.println("3. Withdraw Money");
            System.out.println("4. Transaction History");
            System.out.println("5. Exit");
            System.out.print("Choose an option: ");

            try {
                int choice = Integer.parseInt(scanner.nextLine());

                switch (choice)
                {
                    case 1:
                        atm.checkBalance();
                        break;
                    case 2:
                        System.out.print("Enter deposit amount: ");
                        double depositAmount = Double.parseDouble(scanner.nextLine());
                        atm.deposit(depositAmount);
                        break;
                    case 3:
                        System.out.print("Enter withdrawal amount: ");
                        double withdrawAmount = Double.parseDouble(scanner.nextLine());
                        atm.withdraw(withdrawAmount);
                        break;
                    case 4:
                        atm.showTransactionHistory();
                        break;
                    case 5:
                        System.out.println("Thank you for using the ATM!");
                        new TransactionLogger(atm.getTransactions()).start();
                        scanner.close();
                        return;
                    default:
                        System.out.println("Invalid choice. Please try again.");
                }
            } catch (NumberFormatException e)
            {
                System.out.println("Invalid input. Please enter a valid number.");
            }
        }
    }
}
```

OUTPUT:-

```
Microsoft Windows [Version 10.0.19045.5608]
(c) Microsoft Corporation. All rights reserved.

D:\java>javac ATMSimulator.java

D:\java>java ATMSimulator

ATM Menu:
1. Check Balance
2. Deposit Money
3. Withdraw Money
4. Transaction History
5. Exit
Choose an option: 1
Current balance: ?5500.0

ATM Menu:
1. Check Balance
2. Deposit Money
3. Withdraw Money
4. Transaction History
5. Exit
Choose an option: 2
Enter deposit amount: 20000
```

```
?20000.0 deposited successfully.

ATM Menu:
1. Check Balance
2. Deposit Money
3. Withdraw Money
4. Transaction History
5. Exit
Choose an option: 2
Enter deposit amount: 50000
?50000.0 deposited successfully.

ATM Menu:
1. Check Balance
2. Deposit Money
3. Withdraw Money
4. Transaction History
5. Exit
Choose an option: 3
Enter withdrawal amount: 6000
?6000.0 withdrawn successfully.

ATM Menu:
1. Check Balance
```

```
ATM Menu:
1. Check Balance
2. Deposit Money
3. Withdraw Money
4. Transaction History
5. Exit
Choose an option: 4

Transaction History:
Deposited ?20000.0
Deposited ?50000.0
Withdrawn ?6000.0

ATM Menu:
1. Check Balance
2. Deposit Money
3. Withdraw Money
4. Transaction History
5. Exit
Choose an option: 5
Thank you for using the ATM!
Transaction log updated.
```